PRESIDENT'S OFFICE

REGIONAL EXAMINATION AND LOCAL GOVERNMENT

KIBAHA TOWN COUNCIL

FORM FOUR PRE- MOCK EXAMINATIONS

032/1 **CHEMISTRY 1**

Time: 3 HOURS. **Year: 2024**

Instructions

1. This paper consists of sections A, B and C with a total of eleven (11) questions.
2. Answer all questions in sections A and B and two (2) questions from section C.
3. Sections A carries sixteen (16) marks, section B fifty four (54) marks and section C thirty (30) marks.
4. Non-programmable calculators may be used.
5. Cellular phones and any unauthorised materials are not allowed in the examination room.
6. Write your Examination Number on every page of your answer booklet(s).
7. The following constants may be used.

   Atomic masses: H= 1, C= 12, O = 16, S= 32, Pb= 207, N=14, Cu=63.5, Mg= 24

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SECTION A (16 Marks)

Answer all questions in this section.

1. For each of the items (i) - (x), choose the most correct answer from the given alternatives and write its letter in the answer sheet provided

i. Which is the correct description of nucleons?

A. Nucleons are neutrons, protons and electrons in the nucleus of an atom.
B. Nucleons are neutrons in the nucleus of an atom.
C. Nucleons are neutrons and protons in the nucleus of an atom.
D. Nucleons are protons in the nucleus of an atom.
E. Nucleons are neutrons and electrons in the nucleus of an atom.

ii. How ions are formed?

A. Non-metal atoms losing electrons from their outermost shells.
B. Metal atoms losing electrons from their innermost shells.
C. Metal atoms losing electrons from their outermost shells.
D. Metal atoms gaining electrons in their outermost shells.
E. Metal ions losing electrons from any of their own shell at any energy level

iii. Iron extraction requires reducing agent for the process to be accomplished. From the given list, recognize the suitable reducing agent required.

A. Calcium oxide
B. Carbon dioxide
C. Carbon monoxide
D. Iron
E. Iron(III) oxide

iv. You have been given the following staffs for preparation of oxygen

(i) Source of heat
(ii) Manganese (IV) dioxide
(iii) Hydrogen peroxide
(iv) Potassium Chlorate

Select the combination that will produce oxygen very fast?

A. (i) and (iii)
B. (ii) and (iv)
C. (ii) and (iii)
D. (iii) and (iv)
E. (i) and (iv)

v. Identify the sets of materials that can cause fire outbreak from the given list.

A. Oxygen, carbon dioxide and fuel
B. Oxygen, heat and fuel
C. Oxygen, heat and foam
D. Oxygen, foam and fuel
E. Oxygen, heat and carbon dioxide

vi. The simplest formula of a compound formed when combining 36 g of magnesium and 14 g of nitrogen is:

A. Mg_3N
B. MgN_2
C. Mg_4N_2
D. Mg_2N_2
E. Mg_3N_2

vii. Consider the given trends in physical properties of elements in the Periodic Table.

(i) Electronegativity increase from left to right.

(ii) Densities increase down the group.

(iii) Melting point of metals increases down the group.

(iv) Metallic character increase from left to right.

Which combination demonstrates correct trends?

A. (iii) and (iv)

B. (i) and (iii)

C. (ii) and (iii)

D. (i) and (ii)

E. (i) and (iv)

viii. When sand and wax are put in the same container and some volume of petrol is added, after few minutes only sand is seen. Why wax is not seen after the addition of petrol?

A. Because wax is soluble in petrol.

B. Because wax is insoluble in petrol in petrol.

C. Because the mixture reached at the equilibrium.

D. Because petrol is flammable.

E. Because kerosene is polar solvent.

ix. The following substances comprises of one mole, EXCEPT? -

A. 32 g of oxygen molecules.

B. 2 g of hydrogen molecules.

C. 19 g of hydroxonium ions.

D. 48 g of carbon monoxide molecules.

E. 98 g of sulphuric acid.

x. A four student mixed a solution of a certain Iron salt with sodium hydroxide, reddish- brown precipitate was observed. Which substance was produced?

A. Iron (II) oxide

B. Iron (III) hydroxide

C. Iron (II) hydroxide

D. Iron salt

E. Iron metal

2. Match the chemical composition of fire extinguisher in **List A** with the corresponding extinguisher in **List B** by writing the letter of the correct response beside the item number in the answer booklet provided.

LIST A	LIST B
i. The extinguisher composed of bromochlorodifluoro-methane. ii. The extinguisher composed of potassium acetate. iii. The extinguisher composed of monoammonium phosphate with a nitrogen carrier. iv. The extinguisher composed of fine sodium bicarbonate powder pressurized by nitrogen. v. The extinguisher composed of proteins and fluoro-proteins. vi. The extinguisher composed of tap water pressurized by air.	A. Dry chemical extinguisher B. Fire blanket extinguisher C. APW extinguisher D. Carbon dioxide extinguisher E. Halon extinguisher F. ABC extinguisher G. Sand bucket extinguisher H. Wet chemical extinguisher I. Foam extinguisher.

SECTION B(54 MARKS)

Answer all questions

3. There was diarrhoea in Kazimbele village which was thought to be caused by some Contamination of water with a certain bacteria, as form four student explain six scientific procedures which you will used to investigate such problem systematically.

4. Explain each of the following
   i. How Lead sulphate can be removed from the solution of Sodium sulphate.
   ii. Melting of ice is regarded as a physical change.
   iii. Rusting of iron is regarded as a chemical change
   iv. Sulphur is a non-metal.
   v. Blue colour disappears during electrolysis of copper (II)sulphate solutions when carbon electrode is used.
   vi. Concentrated sulphuric acid is not an electrolyte.

5. (a) Different salts behave differently when heated. Use balanced chemical equations show how most reactive and moderate reactive metal carbonates and nitrates behave when subjected to heat.
   (b) A form four student was asked to react phosphate ion and sodium ion forming compound Q. Suggest the IUPAC name of Q and find the oxidation state of phosphorus in Q.
   (c) Calculate the percentage composition of lead in the Lead nitrate compound.

6. (a) Briefly explain the concept of dilution and dilution factor.

   b) What volume of 10M acetic acid is used to prepare 10ML of 0.5M acetic acid?

7. (a) A steady current of 2A was passed through a solution containing ions of metal (X^{2+}) for 9 minutes. The mass of metal X that was liberated were 0.3552 g. Calculate the relative atomic mass of metal X; and give the name of an element.

   (b) Consider the following reactions:

   $Cu^{2+}_{(aq)} + 2e^- \rightarrow Cu_{(s)}$

   $Zn_{(s)} \rightarrow Zn^{2+}_{(aq)} + 2e^-$

   With reasons;

   (i) Identify the reduction process
   (ii) Identify oxidation process
   (iii) Name the reducing agent
   (iv) Point out the oxidizing agent

8. 25 cm^3 of 0.098M sodium carbonate solution was titrated against hydrochloric acid by using Methyl orange indicator. The following results were obtained

Volume Reading	**Pilot**	**I**	**II**	**III**
Final (Cm^3)	21.05	20.25	40.75	20.55
Initial (Cm^3)	0.00	0.00	20.25	0.00
Used (Cm^3)				

Questions

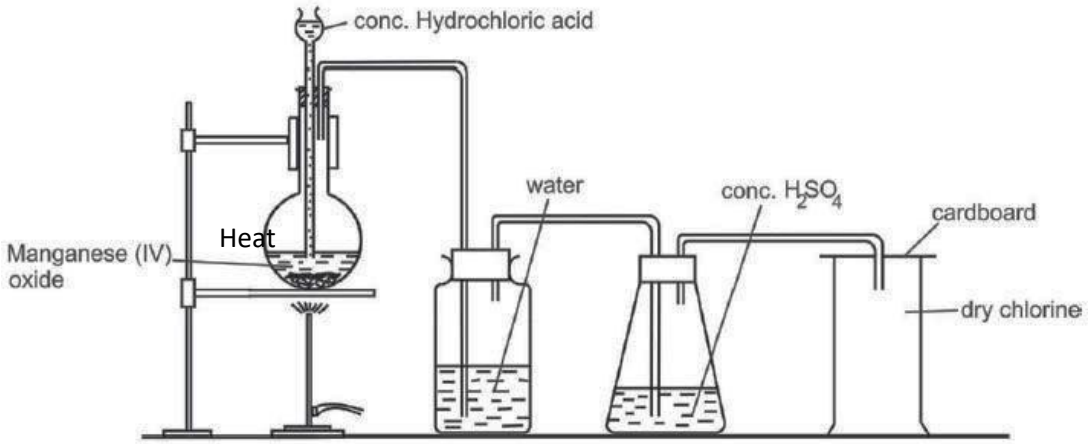
I. Methyl orange colour change from ……..…… to ……..
II. Find the mean titre volume of acid
III. Write balanced equation for reaction
IV. Find the number of moles of alkali used in the reaction
V. Calculate the number of moles of acid used in the reaction
VI. Compute concentration of the hydrochloric acid in moles per dm^3

SECTION C(Marks)

Answer two(2) questions in this section.

9. Lack of safe and clean water for domestic and industrial uses is a serious problem in most urban areas in Tanzania. The major cause of this problem is pollution in the water sources which causes the hardness of water. Explain five methods that could make such water clean and safe.
10. Describe the extraction of iron from its mostly used ore, and write all the Chemical equations for the reactions involved in each stage of extraction.

11. This set up below was used to prepare gas called chlorine in the laboratory. Study it carefully then answer the questions below it:



a) With reference of the set up above, write a balanced reaction equation for the oxidation of concentrated hydrochloric acid.
b) During preparation of chlorine in the laboratory, the gas is passed through water and concentrated Sulphuric acid before collection. State the role of:
   i. Water
   ii. Concentrated Sulphuric acid
c) Why Chlorine gas is collected by downward delivery?
d) State three industrial uses of Sulphuric acid.